Autonomous Agents and Multiagent Systems

A series of notes on the course AAaMS as taught in Politecnico di Milano by Francesco Amigoni and someone else

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Negotiation Part 2 - 11/10

We now focus on the agreement found by the agent, not on the policy.

We'll define a protocol, see how the agents behave in it, and the study the possible outcome that will be reached by the agent.

Let's start with the first protocol:

Rubinstein's Alternating Offer Protocol

I make an offer, the other agent makes a counter offer, I make a new offer, and so on, until we find some agreement.

We assume that we have 2 agents, a, b.

We assume there is one issue over which the agents are negotiating. It can be though as an object that can be divided by the 2 agents.
Let's think of a cake. It can be divided by the 2 agents. We'll call the piece of cake that goes to the agent a x^a. x^b is the part of the cake that goes to b.

we assume that $x^a + x^b = 1$ (it forms the complete cake).

The set of outcomes is trivially the set of all the pairs (x^a, x^b) such that $x^a + x^b = 1$.

in math: $O = (x^a, x^b) | x^a + x^b = 1$ where O stands the set of possible Outcomes.

We assume that time is evolving in discrete period: $t = 1, 2, \ldots$.

We assume that players play in turn, with player a playing first:
So agents a makes offers in odd timestamps, while b makes offers in even timestamps. This is the core of "Alternating" Offer protocol.

the ones above are all and only the assumptions that the agents must do.

Let's talk about the utility of a: it depends on how big the cake is and on the timestamp in which the offer is agreed.

$$U^a(x^a, t) = x^a \cdot \delta_a^{t-1}$$

where δ_a is the discount factor and $0 \leq \delta_a \leq 1$

$\delta = 1 \rightarrow$ the agent is very patient.

$\delta = 0 \rightarrow$ the agent must accept the first offer otherwise it gets zero.

and obviously:

$$U^b(x^b, t) = x^b \cdot \delta_b^{t-1}$$

Rubinstein shows that it is possible to compute a subgame perfect equilibrium:

$$x^a = \frac{1 - \delta_b}{1 - \delta_a \delta_b}$$

$$x^b = \frac{\delta_b}{1 - \delta_a \delta_b}$$

It means that the agent will not find convenient to deviate from these values.

we notice that if

$$\delta_a = 0; \delta_b = 1 \rightarrow x_a = 0;\ x_b = 1$$

$$\delta_a = 1; \delta_b = 0 \rightarrow x_a = 1;\ x_b = 0$$

$$\delta_a = 0.5; \delta_b = 0.5 \rightarrow x_a = \frac{2}{3};\ x_b = \frac{1}{3}$$

in the last case, the agent a gets more cake because it started first.

The cases above have been computed considering that the negotiation can go on forever, but what happens if we have a deadline $t = n$? (n is the last timestamp in which an offer can be done).

Let's suppose even that $\delta_a = \delta_b = \delta$

$n = 1 \rightarrow$ agent b will just accept, can't do anything else. So agent a will propose $offer(1, 0)$ more precisely:

$$\text{STRAT-A}(1) = \text{OFFER}(1, 0)$$
$$\text{STRAT-B}(1) = ACCEPT$$

$n = 2$

$$\text{STRAT-A}(1) = \text{OFFER}(1 - \delta, \delta)$$
$$\text{STRAT-A}(1) = \text{ACCEPT}$$

which translates into:

$$U^a(1 - \delta, 1) = 1 - \delta$$
$$U^b(\delta, 1) = \delta$$

let's revise what we have just written:
The agent a prefers to offer to b δ part of the cake because the utility of b can't be more than δ (in fact $U^b(1, 2) = \delta$), because this way a gets at least $1 - \delta$ as utility, so it's a win win.

let's generalize:

deadline n, $\delta_a = \delta_b = \delta$

$$\text{STRAT-A}(n) = \begin{cases} \text{if a's turn: OFFER(1,0)} \\ \text{if b's turn: ACCEPT} \end{cases}$$

$$U^a(n) = \begin{cases} \delta^{n-1} & \text{if it's a's turn} \\ 0 & \text{if it's b's turn} \end{cases}$$

$t < n$

$$\text{STRAT-A}(t) = \begin{cases} \text{if a's turn: OFFER}(1 - \delta x^b(t+1), \delta x^b(t+1)) \\ \text{if b's turn: if } U^a(x^a, t) \geq UA(t+1) \text{ then ACCEPT else REJECT} \end{cases}$$

Monotonic Concession Protocol

- agents a, b ; U^a, U^b
- O
- simultaneous offer protocol
- $t = 1, 2, \ldots$

From the point of view of agent a:

1. $x^{(a)} \leftarrow \arg\max_{x \in O} U^a(x)$

2. propose $x^{(a)}$

3. receive $x^{(b)}$

4. if $U^a(x^{(b)} \geq U^a(x^{(a)}))$ then ACCEPT $x^{(b)}$ and RETURN
   else $x^{(a)} \leftarrow x \in O$ such that $U^b(x) \geq U^b(x^{(a)})$ and $U^{(a)} \geq 0$

5. goto 2

This algorithm can be applied only when you know the utility of the other agent, which does not happen very often in the real world...

Moreover, the converge of this algorithm can b very long.

Another problem: It can happen that agent a accepts the offer of agent b and agent a accepts the offer of agent b, but they are not the same! we have two different agreements: we need to introduce some tie-breaking to decide the agreement. we can pickup just one at random , or we can find the middle point of the offer and come up with a single agreement.

There is a variant of this protocol that is called the Zeuthen Strategy

Zeuthen Strategy

This is the point of view of a

$1.\ x^{(a)} \leftarrow \arg\max_{x \in O} U^a(x)$

$2.\ \text{propose } x^{(a)}$

$3.\ \text{receive } x^{(b)}$

$4.\ \text{if } U^a(x^{(b)} \geq U^a(x^{(a)})) \text{ then ACCEPT } x^{(b)} \text{ and RETURN}$

$5.\ \text{risk}_a \leftarrow \frac{U^a(x^{(a)}) - U^a(x^{(b)})}{U^a(x^{(a)})}; \text{risk}_b \leftarrow \frac{U^b(x^{(b)}) - U^b(x^{(a)})}{U^b(x^{(b)})}$

$6.\ \text{if risk}_a < \text{risk}_b \text{ then } x^{(a)} \leftarrow x \in O \text{ such that risk}_a > \text{risk}_b$

$7.\ \text{goto } 2$

Property: the final agreement found via this algorithm is a pareto optimal agreement, which means that there is no another agreement that is not worse for one agent and strictly better for the other one.

it finds an agreements which is a Nash BARGAININ solution. It is an agreement that maximizes the product of the two utilities.

Computational Social Choice - 18/10

Applications:

- Aggregation of preferences.
- Formation of coalitions.
- Sometimes you have multiple robots that have to keep a formation, for example military patterns.
  It is based on the fact that there is a leader that is moving and all the other robot are keeping a fixed distance and angle from him. They should also avoid to collide with each other.
  What happens if the leader fails? In this case there is a kind of voting for choosing the new leader, and this is the social choice.

Let's start from the model that we will adopt for studying computational social choice.

we'll have:

- a number of ***agents*** $N = \{1, 2..., n\}$
- a set of **alternatives**, also called **candidates** (in the case of political elections for example) U. we assume $|U| \geq 2$.
- ***preferences***, denoted by the symbol *preference relation* $>_{\sim i}$ (it is an ordering over the alternatives).
  for it being a proper ordering we need to constraint such relations in the following way. we need to say that such relation is
  - complete: $\forall a, b \in U : a >_{\sim i} b \text{ or } b >_{\sim i} a$ (the or is not generally exclusive)
  - transitive: $\forall a, b, c \in U : \text{if } a >_{\sim i} b \text{ and } b >_{\sim i} c \text{ then } a >_{\sim i} c$

Observations: agents and candidates could be disjoint or not depending on the case. In political election agents are people and politicians, while candidates are politicians.

Now let's introduce some symbols:

- ***preference profile*** $R = (>_{\sim 1}, >_{\sim 2}, >_{\sim n})$ (it is an array of preferences, one for each agent).
  a preference can be structured as a pair of agent, candidate. So a preference profile can be represented as a list of pairs.
- a ***set*** of all possible ***preference relations over*** $U : \mathscr{R}(U)$
  $>_{\sim i} \in \mathscr{R}(U)$

Mathematicians game:
We are at a party in which people can drink only one beverage.
6 mathematicians want to drink, in preference order, milk,wine,beer.
5 mathematicians want to drink, in preference order, beer,wine,milk.
4 mathematicians want to drink, in preference order, wine, beer,milk.
So:

- we have 15 agents: $n = 15$.
- we have 3 alternatives: $|U| = 3$
- for the mathematician number 1, who belongs to the 6 mathematicians cited first, we'll have $\text{milk} >_{\sim 1} \text{wine} >_{\sim 1} \text{beer}$. and so on for all the mathematicians.

It is not easy to aggregate together different preference relations with a global choice that is satisfactory for all people. It depends even on the mechanism of voting.

Social Welfare Function

It is a function that given a preference profile returns a preference relation.

$f : \mathscr{R}(U)^n \to \mathscr{R}(U)$

$f(R) \ = \ >_{\sim}$

This is what the function should do, there are different ways to implement it.

We want this function to have the following properties

- <u>Pareto Optimality</u>
  Informal definition: assume that every agent thinks that candidate a is better than b. then it is reasonable to assume that a will be better than b even in the global relation.
  Formal definition:
  $\text{if } a >_i b \ \text{ then } a > b$
  the one above has been written assuming the following: $a >_{\sim i} b \text{ but not } b >_{\sim i} a$, which is not the general case.
  the consequence assumes instead that: $a >_{\sim} b \text{ but not } b >_{\sim} a$.

I had to write the formal definition this way because the actual formal definition would be complex to be written, so Amigoni preferred specifying the assumptions afterwards.

- Independence of irrelevant alternatives (IIA)
  imagine this starting situation

$$f\begin{cases}\ldots >_{\sim 1} a >_{\sim 1} \ldots >_{\sim 1} b >_{\sim 1} \ldots \\ \ldots >_{\sim 2} a >_{\sim 2} \ldots >_{\sim 2} b >_{\sim 2} \ldots \\ \ldots >_{\sim 3} a >_{\sim 3} \ldots >_{\sim 3} b >_{\sim 3} \ldots\end{cases} \rightarrow \ldots >_{\sim} a >_{\sim} \ldots >_{\sim} b >_{\sim} \ldots$$

  In a nutshell we want to say that if you swap the red and yellow elements, and, as we can see, a and b have the same ordering over all the agents, and in case a and b have anyways the same order in the global relation, we can say that our formulation has IIA.

  In case of IIA the ordering of a and b in the global relation does not dependent on how it is called the yellow element.
  Formally:
  $\text{if } R|\{a,b\} = R'|\{a,b\} \text{ then } >_{\sim} |\{a,b\} =>'_{\sim} |\{a,b\} \text{ where } f(R) =>_{\sim} \text{ and } f(R') =>'_{\sim}$

- Non-Dictatorship
  A good social welfare function is such that there is no such an agent which imposes the ordering over the other agents. This is Non-Dictatorship.
  Formally:
  $\text{there is no } i \in N \text{ such that } f(R) =>_{\sim i} \forall >_{\sim i}$

Arrow's Theorem

There is no any social welfare function that satisfies the 3 properties above when you have at least 3 candidates.

The property that is mostly dropped is the IIA.

Social Choice Function

The Social Welfare Function can be too powerful in many situations. take for example the problem of electing 3 new members in parliament over 10 possible candidates. you just need to know the 3 most voted ones, you don't care about the others.

So now we'll define the social choice function

$f : \mathscr{R}(U)^n \times \mathscr{F}(U) \rightarrow \mathscr{F}(U)$

where $\mathscr{F}$ is the powerset of U: $\mathscr{F} = 2^U$, which is the set of all feasible sets of alternatives.

$A \in \mathscr{F}(U)\ A \subseteq U$

$f(R, A) = A' \qquad A' \subseteq A$

Voting Rule

one special case of the social choice function is the ***voting rule***.

Given a preference profile it returns a set of candidates.

$f : \mathscr{R}(U)^n \rightarrow \mathscr{F}(U)$
$f(R) \rightarrow A'$

f is resolute when $|f(R)| = 1$

Computational Social Choice - 22/10

Example

6	5	4
milk	beer	wine
wine	wine	beer
beer	milk	milk

scoring rules: rule that gives some point to each alternative according to the ranking related to each agent.

$$>_\sim (s_1, s_2, \ldots, s_m) \qquad m = |U| \qquad s_i \in \mathbb{R}$$

where s_i is the score of the first alternative in the ranking.

$$s_1 \geq s_2 \geq \ldots \geq s_m \qquad s_1 > s_m$$

Scoring Rules

- Plurality Voting
  Everyone votes for his preferred alternative. for example, milk gets 6 votes, beer 5 votes, and wine 4.
  We are considering one point for each vote.
  $(1, 0, 0, \ldots, 0)$ you consider the first ranked $\rightarrow$ milk
- Borda Count
  $(m-1, m-2, \ldots, 0)$
  $m =$ number of alternatives. In our example we have:
  milk: $6 \cdot 2 + 5 \cdot 0 + 4 \cdot 0 = 12$
  wine: $6 \cdot 1 + 5 \cdot 1 + 4 \cdot 2 = 19$
  beer: $6 \cdot 0 + 5 \cdot 2 + 4 \cdot 1 = 14$. shortly we have that the last position counts 0, the second one counts 1, and the first one counts 2.

Which one between Borda and Plurality can be considered also social welfare function?
Which means: which one will return a ranking?
Borda returns a global ranking among preferences.

Pile Wise Majority Voting

A ***Condorcet Winner*** is an alternatives that wins against every other alternative in a pile wise majority voting.

Pile wise majority voting: a voting where you have only 2 alternatives and the alternatives that get more points is the winner.

A Condorcet winner is a very strong candidate because it wins against a every other candidate.

Do we have a Condorcet winner in the beverage example? (we are considering Borda count)

$milk - beer$

$milk - wine$

$wine - beer$

Yes, wine, because it wins against every other candidate.

A ***Condorcet Extension*** is a voting rule that selects a Condorcet winner if a Condorcet winner exists.

It is not the case of plurality, it may be the case of Borda considering the result but it's actually not:
No scoring rule is guaranteed to select the Condorcet winner if the Condorcet winner exists, no scoring rule is a Condorcet Extension.

Example

6	**3**	**4**	**4**
a	c	b	b
b	a	a	c
c	b	c	a

17 agents in total. 3 candidates.

is there any condorcet winner?

$a - b$ (9 over 8)

$a - c$ (10 over 7)

at this point we know a is a condorcet winner.

$b - c$

Let's see that no scoring rule would have selected a:

these are the generical weights associated with preference ranks: $(1, x, 0)$

$a : 6 \cdot 1 + 3 \cdot x + 4 \cdot 0 = 6 + 7x$
$b : 6 \cdot x + b \cdot 0 + +4 \cdot 1 + 4 \cdot 1 = 8 + 6x$
$c : 0 + 0 + 3 + 4 \cdot x = 3 + 4x$

There is no legal (between 0 and 1) value of x such that $6 + 7x > 8 + 6x$, in fact we obtain that $x > 2$ while we would like an x such that $0 < x < 1$.

Copeland's Rule

In order to determine the winner you give to each alternative one point every time it is winning against another alternative, you give $1/2$ when there is a tie, 0 points when it loses.
$a : 2$
$b : 1$
$c : 0$

Plurality with Runoff

So far we considered voting rules that have a single step.
There are multi step voting rules:
In voting rules that evolve in steps, the winner is found after several steps. An example is plurality with runoff:

Plurality with Runoff: typical in political elections.
It is a 2 stage voting.
In the first stage you run a plurality voting.
Then you take the 2 most voted candidates and run a pairwise majority voting between the 2 (a

ballot).

Example

6	5	4
milk	beer	wine
wine	wine	beer
beer	milk	milk

after 1st stage:

milk	beer
6	5

after 2nd stage:

beer wins over milk (9 over 6):

Conclusion:

Beer wins.

Single Transferable Vote (STV)

Run a plurality voting, then the alternatives that are ranked first by the lowest number of agents are eliminated, then you run a new plurality voting without considering the eliminated candidates.
N candidates $\rightarrow N-1$ round to perform.

Example

6	5	4
milk	beer	wine
wine	wine	beer
beer	milk	milk

I eliminate wine.

6	5	4
milk	beer	beer
beer	milk	milk

beer wins over milk (9 over 6):

I eliminate milk.

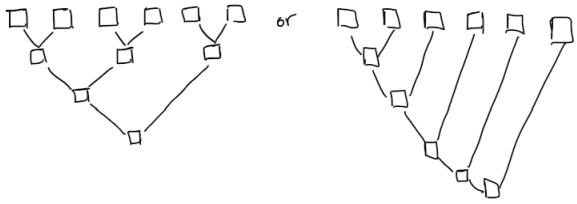
Beer wins.

Is it a condorcet winner extension? no.

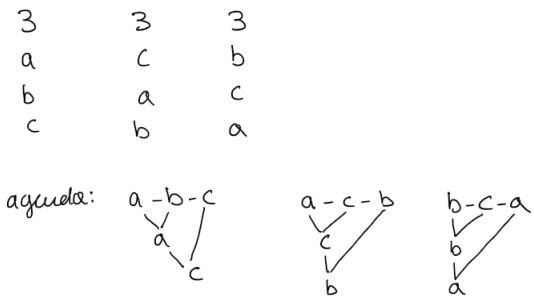
It is not trivial to find a system that selects a good candidate.

Sequential Majority Voting

Combine in different ways a sequence of majority voting.



Agenda : the order in which alternatives are paired in a sequence of pair wise majority voting.



what we are saying in the second sequential majority voting example is:
let's compare a with b first, and then let's compare the winner with the other one. according to different ways to start there are different final outcomes! So whenever someone says "let's start from evaluating those two" there is something wrong with this kind of reasoning.

According to how we select the agenda we have different winners.
Selecting the agenda is not neutral.

Sequential Majority Voting is a condorcet extension because the condorcet winner can never loose in a pair wise.

Strategic Manipulation of vote

Let's consider that we have 4 candidates

1	**2**	**2**	**2**
a	a	b	c
b	c	a	b
c	b	c	d
d	d	a	a

I'm running plurality voting. who is the winner? candidate a because he gets more votes.

Now let's suppose I'm one of the last two players (fourth column).
I'm not happy with the outcomes.
If I know the preference relation of the other candidates what can I do?
I say: if I vote c the winner bill be a. let's vote b, even if it is not my favorite candidate, but it will win, and I like him way more than a. This is called ***strategic manipulation***.

any agent can report wrong information in order to get an advantage

What happens if, in the same example, instead of using plurality voting we use Borda count?

$a : 9$
$b : 14$
$c : 13$
$d : 6$

b wins.

Also Borda count is prone to strategic manipulation..

Gibbard–Satterthwaite Theorem

Any voting system is prone to strategic manipulation.

So you can not avoid strategic manipulation, but you can make it harder to happen.

Probabilistic Voting Rule

So far we have considered deterministic voting systems. The winner is deterministically selected.
It is also possible to use probabilistic voting rules:
The rule is not selecting a candidate, but it is giving in output a probability distribution over the candidates. Then you pick up the winner randomly selecting the candidate according to the probability distribution.
So deterministic voting rule will output something like: the winner is a with 100% probability.
Probabilistic voting rules will output something like: the winner is a with 70%, b with 10% and so on.

Another Gibbard-Satterthwaite Theorem

when

- *there are more than 2 alternatives*
- *you use a probabilistic voting rule*
- *you impose that your probabilistic voting system is Pareto optimal*
- *and it is no manipulable*

then the only probabilistic voting system that is satisfying it is probabilistic dictatorship.

Auctions - 25/10

Auctions ("aste" in italian) are very used in the real world, for example:

- in the stock market
- public concourse

An auction is made out of an auctioneer (usually the one who sells) and some agents (usually the ones who buy). roles could be the opposite, for example for public concourses, in that case the auctioneer is the one who buys.

When you set up an auction you have to set up the rules of the auction. for example if you are the owner of Ebay you have to say what is the minimum and maximum price or which an user can sell an item.
You have to set even the rules: for example, is it possible to propose a price and get immediately the item? or we must wait for the end of the auction?

What an agent decides to do is called *strategy*.

Different kind of good's auctions

- single-good
  you make an offer only on one item
  (i.e. Van Gogh's painting auction)
- multi-good
  you make an offer on some of a set of goods, or all of them, or just one.
  (i.e. telecommunication auctions)

if you are the buyer you can fix a maximum price that you are willing to offer.
At the same time, if you are the seller, you can fix a minimum price that you are willing to accept.
This last price is called ***reservation price***.

Canonical Auctions

- English Auctions
  The auctioneer announces an initial price (the reservation price).
  The agent can make a new offer with the following constraint: the new offer should be larger than the largest offer made so far by any of the other agents.
  When the auction in this case ends? when there is no new offer.
  The agent will pay exactly the amount of money that he has offered.
  - offer made by agents
  - usually big jump of prices
  - the auctioneer is more passive. he has no control
- Japanese Auctions
  At the beginning all the agents will be standing up. The auctioneer will start from his reservation price and he will call increasing prices.
  If an agent sits down he is out of the auction forever.
  The winner is the last standing agent.
  The amount of money he will pay is the price that the auctioneer has proposed at last.
  What happens if we have two agents standing and they sit down at the same time?
  One tie-breaking mechanism is: a new auction is run just for those agents.
  - offer made by auctioneer.
  - more linear.
  - the auctioneer is more in control of the situation.
- Dutch Auctions
  there is a clock. it does not show the time but the price.
  it starts from a really high price and goes down.
  the auction ends when one of the agents stops the clock.
  The winner is the agent that stopped the clock and the amount of money he will pay is the price that is displayed by the clock at time of stopping.
  (i.e. for selling flowers, for selling fish. it is really fast. used for selling thinks that are not of a big value and you have to sell them fast.
  the auctioneer can set the speed of the clock)
- Sealed-bid Auctions
  the agents are sending their bids (offer) to the auctioneer using a sealed envelope.
  There are several families of sealed-bid auctions, but we'll see just two:
  - ***First Price***
    The auctioneer selects the largest possible offer.
    The agent that made the best offer will get the item and pay exactly the amount he has offered.

- ***Second Price*** (also called Vickrey auction)
  The auctioneer selects the largest possible offer.
  the winner is the one that made the largest offer but he does not pay his offer, but the second highest.

(used for allocating public works/concourses). Sealed-Bid auctions are *asynchronous*! no need to have all the agents present at the same time in a specific place.

Common value & IPV

if in an auction I'm selling an 1 euro coin, would you pay more than 1 euro for it? no, if it has no special feature. all the agents agree on a price, that is the ***common value***.
If such coin was instead a very very rare one? it would have a higher value.

Another example: I'm going to buy a house. I've to consider many characteristic depending on my needs (if I'm in a wheelchair I will value it more if it develops on just one floor).
It has a different value depending on the person that is going to buy it.
In the case above we'll then assume that goods have ***independent private values*** (IPV).

Important Considerations

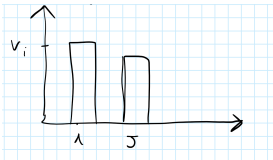
Considerations on Second Price Sealed-Bid Auctions

Now let's do some considerations on Second Price Sealed-Bid Auction:

If I'm using Second Price Sealed-Bid Auction, offering your true evaluation of the object is the dominant strategy (if you think that the object is worth 50, you offer 50).

why? here it is:

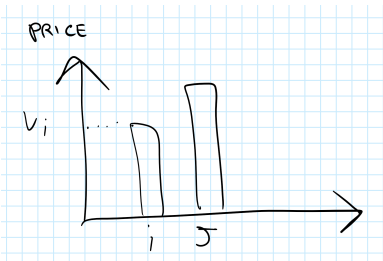
Case 1:
Let's assume that v_i is going to win this auction.
the y axis represents the price paid.



is it wise for i to propose a higher offer? yes, he would still pay the amount offered by j.
Is it wise for i to propose a lower offer? no, he could lose the item.

Case 2:

Let's assume that v_j is going to win this auction.



is it wise for i to propose a higher offer? no because if you offer more than v_i you could end up with gaining the item, but paying more than what you value it (you value it v_i), which is not rational.
is it wise for i to propose a lower offer? no because it would reduce his chances of getting the item.
This is a mechanism that enforces the agents to tell the truth!!

Considerations on English and Japanese Auctions

the winning strategy is to offer an increase until my true evaluation.

Considerations on Dutch and First Price Sealed-Bid Auctions

There is no dominant strategies in this case.

There are though some interesting results:

- strategical equivalence
- equilibrium strategies
- risk neutrality

*Dutch and FPSB are **strategically equivalent**.*

Why?

let's suppose there are 4 agents and I'm involved in a Dutch auction. the price starts at 100 and it reaches 80, which is my true value for the item. should I stop or not?
If I don't know how much the other agents are willing to pay, it could be that it is not the best thing to do to stop the evaluation, because I could end up paying less for the item.
the same happens with FPSB (if I know that the second highest offered value is 70, I can offer 71, even if my true value is 80).

In this cases we can have some ***equilibrium strategies***: if an agent deviates from his strategy and the others are not changing theirs, such agent is going to lose (same thing as Nash Equilibrium).

If you have n agents, the following is an equilibrium strategy:

$$(\frac{n-1}{n}v_1 \frac{n-1}{n}v_2, \ldots, \frac{n-1}{n}v_n)$$

where v_i is the true value for agent i (the value he is willing to pay).

Moreover we can say that agents are ***risk neutral***:
Let's explain it via an example.
I have 0 euro in my pockets. you give me a bill of 10 euros. my utility was zero, now it is 10, good.
now let's instead assume I have a million euros in my pocket. you give me 10 euros. the increment is exactly the same in both cases, but in the first case I value more passing from 0 to 10, then from 1 million to 1 million and ten. The scenario just described is *not* risk-neutral.
I'm risk neutral if I value in the same way the increment, independent on how much I have.

Auctioneer's Point of View

Let's now talk about the auctioneer:

If I'm going to set up an auction, what mechanism should I choose?

Revenue Equivalence Theorem

n risk-neutral agents.

IPV.

the values v_i are uniformly taken from an interval that is $[\ \underline{v}\ ,\ \overline{v}\ ]$.

if

- the good is allocated to the agent that made the largest evaluation (not the largest offer!)
- the agent with the smallest evaluation has basically no chance to get the good.
  (it's expected utility is set to zero)

then

- any kind of auction that satisfies such preconditions, the provide to the auctioneer the same expected revenue

this means that all the auctions that e have seen satisfy the preconditions, so all of them are equivalent from the point of view of the auctioneer.

things to think about for next lecture

- take the English auction, try to verify that the good is always allocated to the agent with the largest v_i.
- try to think what is the amount of info that agents participating in auction are disclosing to the others in the different auctions we have seen.

this will show me why sealed bit auctions are very popular in real world scenarios.

Multiple Items Auctions - 29/10

In this scenario we have that multiple items are auctioned at the same time.

Multiple good/item auctions are combinatorial auctions, they are called this way because we use a combinational approach to solve them.

Example: A company selling stocks in blocks $\rightarrow$ you can make an offer for the first and third stocks block, or for the first and the second one.

the problem is described with the followings:

- **Agents** $N = 1, 2, 3, \ldots, n$
- **Goods** $G = 1, 2, 3, \ldots, m$
- **Valuation function**
  The domain is the power set (set of all possible subsets) of the goods.
  The codomain is a real number.
  To every subset of goods is associated a real number.

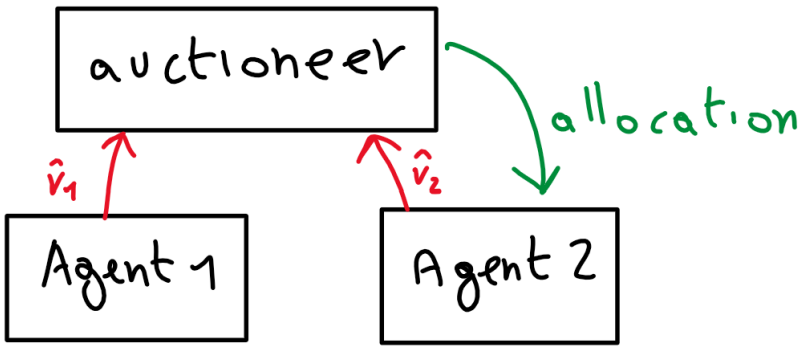
$$v_i : 2^G \rightarrow \mathbb{R}$$

- **Submitted bid of the auctioneer**

$$\hat{v}_i$$

- $v_i - \hat{v}_i$
- Agents can decide not to make an offer for one or a group of goods.
- Agents can offer less than the real value

Extension to the combinatorial auction



Important difference: this time the bid is not a number, it is a function.
The auctioneer will receive all the bids and will decide which agent is the winner $\rightarrow$ difficult to do.

Complementarity

The evaluation function can express some sort of complementarity when the value of the union between two disjoint subsets of G is larger than the sum of the two.
If it holds for every subset it means that agent i thinks that having goods together is better than having it alone.
That means that the two objects are complementary. (Right and left shoes for example).

$$T, S \subseteq G \quad T$$

EXCETERA, TO BE COMPLETED

Lecture 8/11

To be done, wait for it.

Computational Coalition Formulation - 12/11 & 15/11

We are talking about cooperative games:
agents are playing games in which there are two big differences with respect to classical games that are studied in game theory:

- there is some benefit in cooperating
- if agents make some agreement, than this agreement is binding (there are ways outside of the game for enforcing this agreement)

In this kind of games, actions are taken by groups of agents.

We are going to look to a specific kind of cooperative game, called ***Transferable Utility Games***: *the utility is not individual, but given to the group of agents in the coalition* (such coalition can be then divided within the coalition).

We are going to introduce the concept of ***characteristic function games***: *as a group my utility depends only on what I do, not on what other groups do*.

Characteristic Function & TU Games

$$A = \{a_1, a_2, \dots, a_n\} \qquad \text{are agents}$$

$$v : 2^A \to \mathbb{R} \qquad \text{is our characteristic function}$$

such function is called characteristic function and associated a real number to a specific coalition.

assumptions:

$$v(\{\}) = 0$$
$$v(C) \geq 0 \qquad \forall C \subseteq A$$

each C is called coalition.

The whole set of agents A is called sometimes *grand coalition*.

Introductory Example

There are 2 children.
Charlie has 6 dollars.
Marcy has 4 dollars.
Patty has 3 dollars.
Charlie wants to buy ice cream which is 500g and it costs 7 dollars. The others are explained mathematically.
$w_c = 500g;\ c_c = 7\$$
$w_m = 750g;\ c_m = 9\$$
$w_p = 1000g;\ c_p = 11\$$

$A = \{c, m, p\}$
Let's write the characteristic function:

$v(\{c\}) = v(\{m\}) = v(\{p\}) = 0$
$v(\{c, m\}) = v(\{c, p\}) = 750$
$v(\{m, p\}) = 500$
$v(\{c, m, p\}) = 1000$

We don't know though how these children will split the ice cream.
Let's introduce the concept of ***outcome***.
the outcome is a pair $(CS, \bar{x})$ where

- CS is the ***Coalition Structure***: simply a partition of the set of agents.
  $CS = \{c_1, c_2, c_3, \dots, c_k\}$
  $U_{i=1}^{k} c_i = A \quad c_i \cap c_j = \{\}\ \forall 1 \leq i, j \leq j$
- $\bar{x}$ is the ***Payoff Vector***, $\bar{x} = (x_1, x_2, \dots, x_n)$
  $\sum_{i \in C} x_i = v(C) \quad \forall C \in S$

Another example

$v(\{1, 2, 3\}) = 9$
$v(\{4, 5\}) = 4$
$(\{1, 2, 3\}, \{4, 5\}), (3, 3, 3, 3, 1))$ where the orange part is the CS and the red one is $\bar{x}$.
This outcome respects the efficiency principle which says that the sum of the individual payoffs should sum the coalition payoff.

Superadditivity

An important property that a TU game can have is superadditivity:

$$v(\{C \cup D\}) \geq v(C) + v(D)$$

It is always convenient, overall, to form the grand coalition.

I said overall because individual utilities could be lower than in other coalitions.
In this case $CS = S = \text{grand coalition}$.

is the game $v(C) = |C|^2$ superadditive? yes it is.

Convex Property

A game is convex when
$v(C \cup D) + v(C \cap D) \geq v(C) + v(D) \;\; \forall C, D \subseteq A$

If a game is convex then it is even superadditive because a superadditive game is a special case of convex game in which the intersection is always empty

Simple Property

$v(C) \in \{0, 1\} \; \forall C \in A$

How to choose the payoff vector?

Let's modify a little the introductory example:

There are 2 children.
Charlie has 4 dollars.
Marcy has 3 dollars.
Patty has 3 dollars.
Charlie wants to buy ice cream which is 500g and it costs 7 dollars. The others are explained mathematically.
$w_c = 500g; \; c_c = 7\$$
$w_m = 750g; \; c_m = 9\$$
$w_p = 1000g; \; c_p = 11\$$

$v(\{c\}) = v(\{m\}) = v(\{p\}) = 0$
$v(\{c, m\}) = v(\{c, p\}) = 500$
$v(\{m, p\}) = 0$
$v(\{c, m, p\}) = 750$

This game is superadditive so we can say that the grand coalition is going to form.

Now let's look how we can decide the payoff vector

$(A, (0, 0, 0))$ is possible? no, not efficient.
$(A, (200, 200, 350))$: the payoff vector is possible but it's not a good one because Charlie and Marcy are going to get less than Patty. Charlie and Marcy can split from Patty and get more each (250 instead of 200).
We say that the above outcome is not ***stable***.
A stable outcome is captured though the concept of ***core***.

Core

$$\text{core} = \{(CS, \bar{x} | \sum_{i \in C} x_i \geq v(C) \;\; \forall C \subseteq A\}$$

The core embeds the idea of stability.
So, given a certain $\bar{x}$ we check whether it is a solution acceptable by everyone.
the payoff $(250, 250, 250)$ is in the core because for example $250 \geq 0$ if we consider $C = \{1\}$.
Even the payoff $(750, 0, 0)$ is stable, but it is not **fair**!
Even $(250, 250, 250)$ is not completely fair because Charlie has more money than the others two so he should get more ice cream.

Example about stability

$A = \{1, 2, 3\}$
$v(C) = \begin{cases} 1 \;\; if|C| \geq 2 \\ 0 \;\; otherwise \end{cases}$

This game is superadditive and it has no solution in the core.
The core can tell you if a payoff vector is stable but it cannot tell you if it is not fair.
Property: *the core of a convex game is never empty*.

Shapley Value

the Shapley value embeds the idea of fairness

Example about fairness

$A = \{1, 2\}$
$v(\{1\}) = v(\{2\}) = 5$
$v(\{1, 2\}) = 20$
The game is superadditive.
A fair outcome is $((\{1, 2\}), (10, 10))$, but even $((\{1, 2\}), (5, 15))$ and many more.

Shapley Value

Let's introduce some terminology:

$\Pi^A = \text{permutations over } A$ (it is an ordering over the agents)

$c_\pi(a_i)\text{the number of the agents that in the ordering p are appearing before agent } a_i$

$\Delta_\pi(a_i) = v(C_\pi(a_i) \cup \{a_i\}) - v(c_\pi(a_i)) = \text{marginal contribution of agent } a_i$

The formula of Shapley value will be introduced next time